

Topics: 14.1 Functions of Several Variables; 14.2 Limits and Continuity; 14.3 Partial Derivatives (Part I)

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1. A scuba air tank is built from a right circular cylinder of radius x and height y , with two hemispherical end caps of radius x attached (one on each end of the cylinder).
 - (a) Express the volume V of the tank as a function of x and y .
 - (b) Evaluate $V(3, 2)$ and explain the meaning of this value.
2. Consider the function $F(x, y, z) = \sqrt{y} - \sqrt{x - 2z}$.
 - (a) Evaluate $F(3, 4, 1)$.
 - (b) Find the domain of F .
3. Find and sketch the domain of the function.
 - (a) $f(x, y) = \frac{\ln(2 - x)}{1 - x^2 - y^2}$
 - (b) $g(x, y) = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x^2 + y^2 - 4}}$
4. Sketch the level curves of the function for the given values of k .
 - (a) $f(x, y) = x^2 - y$, for $k = -2, 0, 3$.
 - (b) $g(x, y) = x^2 + y^2$, for $k = 1, 4, 9, 16$.
5. Evaluate the limit.
 - (a) $\lim_{(x,y) \rightarrow (1,1)} \frac{x^2y^3 - x^3y^2}{x^2 - y^2}$
 - (b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1}$
 - (c) $\lim_{(x,y,z) \rightarrow (1,-1,1)} \frac{2xz + 3x + 2yz + 3y}{x + y}$
6. Show that the limit does not exist.
 - (a) $\lim_{(x,y) \rightarrow (0,0)} \frac{3x - 2y}{4x^2 - y^2}$
 - (b) $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2 + 3y^2}$
7. Determine the set of points at which the function $f(x, y) = \ln(1 - x - y)$ is continuous.
8. Find all first order partial derivatives of the function.
 - (a) $f(x, y) = (x^2y - y^3)^5$
 - (b) $g(x, y, z) = 4x^3y^2 - e^zy^4 + \frac{z^3}{x^2} + 4y - x^5$

(c) $h(x, y, z) = \frac{x \sin y}{z^2}$

9. Consider the function $f(x, y) = y \arcsin(xy)$.

- (a) Find the partial derivatives f_x and f_y .
- (b) Evaluate $f_x(1, \frac{1}{2})$. Interpret this value as a slope.
- (c) Evaluate $f_y(1, \frac{1}{2})$. Interpret this value as a slope.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Functions of Two and Three Variables. A function of two variables is a rule that assigns to each point (x, y) in a subset of $\mathbb{R}^2$ a real number. A function of three variables is a rule that assigns to each point (x, y, z) in a subset of $\mathbb{R}^3$ a real number.

Unless otherwise specified, by the **domain** of $f(x, y)$, we mean the set of all points (x, y) for which the expression for f produces a real number (similarly for functions of three variables).

Level Curves and Level Surfaces. The **level curve** of $f(x, y)$ corresponding to the value k is the curve given by

$$f(x, y) = k.$$

For functions of three variables, the analogous objects are **level surfaces**, given by

$$g(x, y, z) = k.$$

Limits. If $f(x, y)$ can be made arbitrarily close to L by taking (x, y) sufficiently close to (a, b) (but not equal to (a, b)), then we write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L.$$

On the other hand, if $f(x, y)$ does not approach a single value as (x, y) approaches (a, b) , then the above limit does not exist. Analogous definitions apply for functions of three variables.

Partial Derivatives. Let f be a function of two variables, and write $z = f(x, y)$. The partial derivative of f with respect to x is denoted by f_x or $\frac{\partial f}{\partial x}$ and represents the rate of change of z with respect to x when y is fixed. To find f_x , we regard y as a constant and differentiate f with respect to x . Similarly, the partial derivative of f with respect to y is denoted by f_y or $\frac{\partial f}{\partial y}$ and represents the rate of change of z with respect to y when x is fixed. Partial derivatives are also defined similarly for functions of three or more variables.

Suggested Textbook Problems

Section 14.1: 1-56, 61-72

Section 14.2: 1-50

Section 14.3: 1-74